

Beyond the Leaderboard: Counterfactual Diagnosis of End-to-End and VLA Driving Policies Under Domain Shift

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Abstract—End-to-end and vision-language-action (VLA) driving policies are compared by leaderboard rank, but a deployment decision needs to know how a policy will behave at a new site, and why.

Attempting to predict a policy’s performance at a new deployment site from an existing leaderboard therefore works poorly, because a rank reflects only the outcome, not the behaviour behind it. On six released policies, rank on nuScenes open-loop metrics or on NAVSIM’s leaderboard does not carry over to scenes with a pedestrian near the ego corridor at a new site (ρ from -0.54 to $+0.49$).

We propose a counterfactual check-up: a few hundred real frames, each edited two ways (pedestrian removed, or re-rendered at night), every edit verified by an independent detector, and the change in the planned trajectory read as a diagnosis rather than a score. By distilling causal feature axes from these edits, five exams separate what a score merges: how far the policy plans to drive, whether seeing the pedestrian buys safety, whether that response scales with danger, whether the plan moves when nothing requires it, and how much an irrelevant lighting change moves it.

Across 246 near-pedestrian scenes and six policies, among the cells in which the pedestrian does lie on the planned path only 1.9% of responses are genuine avoidance, and on those cells, under our open-loop protocol, no policy’s plan diverts or slows. In a pre-registered test from left- to right-hand drive, orderings and verdicts transfer but point values do not. The exam whose ordering best matches how often a policy keeps clear of the pedestrian on the new side is how steadily it drives ($\rho = 0.94$), not how strongly it reacts. Most verdicts settle within a few dozen frames, while hazard sensitivity needs hundreds. Code and edited frames will be released.

I. Introduction

A leaderboard rank does not tell you how a driving policy will behave in a new domain. End-to-end and vision-language-action (VLA) driving policies are compared by benchmark scores, and a deployment decision is, implicitly, a bet that the rank those scores produce will still hold once the policy is driving at the new site, not just on someone else’s roads. On scenes where a pedestrian is near the ego corridor, that bet fails for the six released policies we test: their rank on nuScenes’ [1] open-loop metrics or on NAVSIM’s [2] leaderboard score does not carry over to the deployment site (rank correlation ρ from -0.54 to 0.49 , Section IV-E), a gap that open-loop scores are known to open [3]–[5]. The reason is not noise but mechanism. A score is an outcome measure, and an outcome can be right for the wrong reason: a policy that never approaches

a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data, because a closed-loop simulator for the target domain or a fleet test is exactly what the decision is meant to save.

No evaluation yet measures how a policy will perform in the target domain, so organisations fall back on testing and post-training in the deployment setting itself, rider-only fleets accumulating tens of millions of miles before a safety claim [6] or a robot policy improved on its own deployments [7], a loop whose cost in real-world miles is prohibitive [8].

Cross-domain work in domain adaptation [9] and generalisation [10] solves the shift through training and does not assess how a policy behaves in the target domain, so it leaves the selection problem at deployment open.

Closer to our setting, [11] improves a VLA’s out-of-distribution generalisation by aligning its visual representations, but it presumes an already-chosen vision-language model and offers no procedure for selecting among policies, vision-only ones included. Representation engineering [12] locates and steers concepts in a language model’s activations, yet it is built around semantic prompts and does not carry over to policies whose input is images. Function vectors [13] show that a task can be read out of, and transferred through, a language model’s hidden states. The same instruments have no counterpart for the cross-domain evaluation of physical-AI policies. A systematic method for that evaluation is still missing.

To address this, we propose the Check-Up, a counterfactual diagnosis in three stages (Fig. 1). (a) We construct pixel-level paired data: every logged frame is queried with and without the target pedestrian and under a night-style photometric perturbation, and an independent detector verifies every edit. (b) From each pair we distil two axes of causal attribution—the faithfulness displacement F , the plan change caused by the pedestrian itself, and the interference displacement I , the change the same policy produces under an edit that carries no driving-relevant information (3). (c) Five exams, each with a control and a reference threshold, are built on these paired plans and axes (Fig. 1c): how far the policy plans to drive,

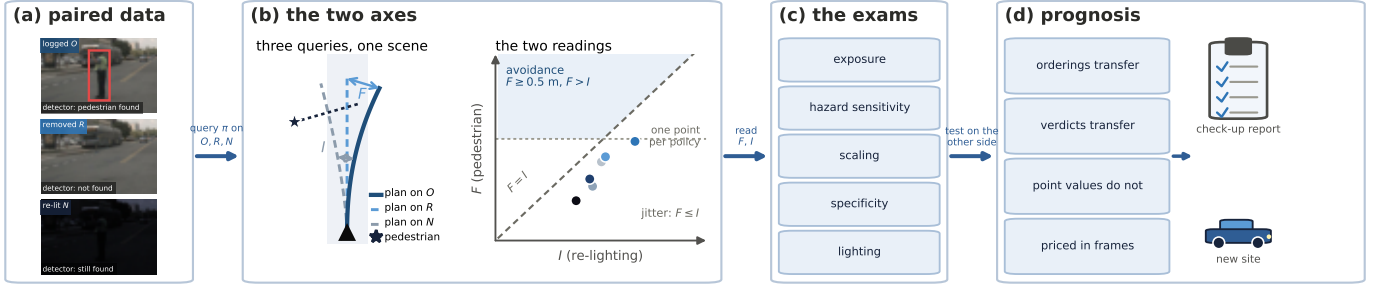


Fig. 1. The Check-Up framework. Pixel-level paired data (a) yield the two causal axes (b) and the five exams built on them (c). Diagnosed on one driving side and tested on the other, they give a priced prognosis, the check-up report (d).

whether seeing the pedestrian buys safety, whether that benefit scales with the hazard, whether the plan moves when nothing requires it, and how much the irrelevant edit moves it. (d) Read on one driving side and tested on the other, the exams yield a prognosis for the target domain: the exam that transfers reproduces the new side’s ordering by how often a policy keeps clear of the pedestrian ($\rho = 0.94$), where the standard scores reach only from -0.54 to 0.49 , and each verdict can be priced in frames before it is used.

The contribution is the Check-Up, a counterfactual diagnosis of end-to-end and VLA driving policies under domain shift, in three parts.

- 1) Pixel-level paired data. Every logged frame is queried with and without the target pedestrian and under a night-style perturbation, each edit verified by an independent detector. Six released policies are read on two driving sides.
- 2) Two causal axes. A response is read as the faithfulness displacement F and the interference displacement I (3), and counts as avoidance only if it moves, exceeds the jitter, and buys distance. Where the pedestrian lies on the planned path, 1.9% of responses qualify, and the lighting edit moves every plan further than the pedestrian does.
- 3) A priced prognosis. Diagnosed on one driving side and tested on the other, orderings and verdicts transfer: the exam that transfers reproduces the new side’s clearance ordering ($\rho = 0.94$), against -0.54 to 0.49 for the standard scores, and each verdict is priced in frames—lighting within 14, exposure within 51.

II. Related Work

a) Driving policies and their benchmarks: We examine bird’s-eye-view (BEV) planners of the TransFuser lineage [14] (LTF, DiffusionDrive [15], DiffusionDriveV2 [16]) and vision–language–action models (SimLingo [17], AutoVLA [18], Alpamayo-1.5 [19], [20]). NAVSIM [2] scores plans by multiplying hard gates with a weighted average of soft terms, open-loop scores are known to reward proxies [4], [21], and closed-loop benchmarks [22] score a policy in simulation. We read released policies on real

frames and show that the six-policy spread is carried by a single gate.

b) Behavioural tests and causal confusion: Checklists [23] and corruption benchmarks [24] probe models beyond aggregate accuracy, but a metric drop does not say whether a policy became more cautious or less. Imitation learners respond to correlates of the expert’s action rather than to its causes [5], [25], [26]. Our exams intervene on a cause (a pedestrian) and a nuisance (lighting) directly and read signed plan changes.

c) Counterfactual editing: Counterfactual explanations edit the scene [27], and CausalAgents [28] deletes non-causal agents to test motion forecasters. Our removal edit applies the same idea at pixel level to a driving policy’s input, and every edit is verified by a detector.

III. The Check-Up: Interventions, Axes and Exams

A. Units, interventions and plans

Write a policy as a map π from a sensory-and-state input to a planned trajectory $X = (X_t)_{t \in [0, 2.5]}$, $X_t \in \mathbb{R}^2$ in the ego frame, sampled at 0.1s. A unit is a pair $u = (f, v)$: a logged frame f and an input ego speed v . We take v from the log for every frame and, wherever the pedestrian’s logged future is available, additionally $v \in \{4, 8\}$ m/s, which places the pedestrian on the planned path in scenes where the logged ego was stationary. The same grid is used on both corpora so that all six policies are read on one unit pool. The readouts are insensitive to it (widening the grid to $\{2, 4, 6, 8\}$ leaves the specificity and scaling orderings identical, $\rho = 1.00$).

Three interventions act on the input, leaving the policy untouched:

$$\begin{aligned} \mathcal{E}_O &= \text{id}, & \mathcal{E}_R &: \text{delete the pedestrian,} \\ \mathcal{E}_N &: \text{re-light at night,} \end{aligned} \quad (1)$$

and yield the paired plans $X^O = \pi(\mathcal{E}_O u)$, $X^R = \pi(\mathcal{E}_R u)$, $X^N = \pi(\mathcal{E}_N u)$. The letters O, R, N name the three conditions only: an intervention carries one as a subscript, a plan as a superscript. All quantities below are functionals of these plans and of the pedestrian’s logged future P_t . The horizon 2.5s is the one all six policies share.

The displacement between two plans is

$$\bar{D}(X, Y) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|X_t - Y_t\|, \quad \mathcal{T} = \{0, 0.1, \dots, 2.5\}, \quad (2)$$

and the two quantities the rest of the paper is written in are

$$F = \bar{D}(X^O, X^R), \quad I = \bar{D}(X^O, X^N). \quad (3)$$

We call F the faithfulness displacement—the plan change attributable to the pedestrian itself—and I the interference displacement—the change the same policy produces under an edit that carries no driving-relevant information, and therefore its own jitter floor. A policy that uses the pedestrian and ignores the lighting has $F \gg I$. The reverse ordering means the plan is driven by something the road does not care about.

a) Realising $\mathcal{E}_R$: The target is localised by projecting its annotated 3D box into the camera rather than by taking the dataset’s 2D boxes, whose association to the tracked object is unreliable for the short-range, partially occluded pedestrians of these scenes. It is segmented with SAM [29] (Fig. 1a) (box and torso prompts, clipped to the box, dilated by 9px) and inpainted with LaMa [30]. For the LiDAR policy the points inside the box are deleted. VLA policies read several cameras and only the front one is edited, so we retain only frames in which the target appears in no other view.

b) Realising $\mathcal{E}_N$: Each image $\mathbf{f} \in [0, 1]^{h \times w \times 3}$ the policy consumes is mapped by a closed-form photometric transform

$$\mathcal{E}_N(\mathbf{f}) = \text{clip}[g(\text{sat}_s \mathbf{f})^\gamma \odot \boldsymbol{\tau} + \sigma \boldsymbol{\epsilon}], \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, 1), \quad (4)$$

with sat_s the saturation scaling, $\boldsymbol{\tau}$ a per-channel tint and the noise seed fixed per frame. This is a photometric perturbation (gamma, gain, saturation, noise), not a captured night scene. We call it the night-style perturbation and use $(\gamma, g, s, \sigma) = (2.2, 0.45, 0.45, 7/255)$ with a blue tint (Fig. 1a).

c) Validity of the interventions: An independent detector (Faster R-CNN [31], COCO, score ≥ 0.5) recovers the target in 100% of the original images, in 8.5% under $\mathcal{E}_R$ and in 89.9% under $\mathcal{E}_N$ (IoU ≥ 0.5): removal suppresses the pedestrian for an observer outside the policy, re-lighting leaves it visible. Pasting inpainting patches on empty road bounds the residual artefact at 5–17% of the removal effect.

B. Readouts

With $c_t(X) = \|X_t - P_t\| - r_e - r_p$ the clearance net of the ego half-width $r_e = 1.0$ m and the pedestrian radius $r_p = 0.4$ m, a plan is summarised by four functionals,

$$S(X) = \text{mean}_t \min(c_t, \bar{c}), \quad A(X) = \mathbb{I}[\min_t c_t > 0], \quad (5)$$

$$C(X) = \text{clip}(\min_t c_t / d_0, 0, 1), \quad T(X) = \frac{1}{2.5} \inf\{t: c_t < d_0\}, \quad (6)$$

with $T = 1$ if the set is empty. The constants are physical, not tuned: r_e is the half-width of the nuScenes ego box (4.08×1.85 m), r_p the disc a standing adult occupies, $d_0 = 1$ m the outer edge of personal space [32], and $\bar{c} = 10$ m caps the contribution of scenes in which the pedestrian is far away. Equations (5)–(6) are the standard surrogate-safety measures [33], read as distance to a person over the whole trajectory. Speed is deliberately not a readout: the same deceleration is safe or unsafe depending on where the person is.

C. Exams

Let $L(X) = \sum_t \|X_{t+\delta} - X_t\|$ denote arc length and H the logged human trajectory.

a) Exposure: $\text{Exp} = \text{med } L(X^O) / L(H)$ over frames with $L(H) \geq 0.5$ m.

b) Hazard sensitivity: A unit needs a reaction when the pedestrian-blind plan enters personal space, $C(X^R) < 1$. On those units, writing $\Delta(\cdot) = (\cdot)(X^O) - (\cdot)(X^R)$,

$$\text{HS} = \frac{1}{4} [\Delta A + \Delta C + \Delta T + \tanh(\Delta S / d_0)] \in [-1, 1]. \quad (7)$$

Each term lies in $[-1, 1]$ by construction and $\tanh$ squashes the single unbounded one at the d_0 scale. The composite is ours, its ingredients are not. Its four terms are the surrogate-safety measures of (5)–(6), and averaging normalised sub-scores into one index is how the benchmark we compare against forms its extended predictive driver model score (EPDMS) [2], keeping the two commensurable. What the check-up adds is that every sub-score is a paired difference across $\mathcal{E}_R$ —the ablation form used to attribute behaviour to an input in interpretability work [12]. HS is read against the model-independent hazard

$$a_{\text{req}} = \frac{v^2}{2 \max(d - 2, 0.5)}, \quad \text{TTC}_0 = \frac{d - 2}{v}, \quad (8)$$

the deceleration required to stop short of a pedestrian at distance d and the corresponding time to collision (TTC) [33]. Units with $v < 1$ m/s are excluded: a stationary ego can move neither toward nor away from anyone.

c) Scaling: $\text{Sc} = \rho(\text{HS}, a_{\text{req}})$ over needed units, with ρ Spearman’s rank correlation.

d) Specificity: On units needing no reaction, the minimal-change principle gives

$$\text{SP} = \exp(-\bar{D}(X^O, X^R) / 0.5 \text{ m}) \in (0, 1], \quad (9)$$

so $\text{SP} \rightarrow 1$ iff an irrelevant edit leaves the plan where it was.

e) Lighting: The causal faithfulness ratio (CFR) weighs the two displacements of (3) against each other,

$$\text{CFR} = \frac{\mathbb{E} \bar{D}(X^O, X^R)}{\mathbb{E} \bar{D}(X^O, X^N)} = \frac{\mathbb{E} F}{\mathbb{E} I}, \quad (10)$$

A policy for which lighting is irrelevant has $\text{CFR} \gg 1$. The thresholds ($\text{HS} \geq 0.15$, $\text{Sc} > 0$, $\text{SP} \geq 0.8$, exposure within $[0.8, 1.2]$, $\text{CFR} > 1$) are reference values we set, not population norms.

IV. Experiments

A. Setup

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B, three cameras $\times$ four frames) and Alpamayo-1.5 (four cameras, a diffusion head over the vision-language model cache), all with released weights. The three BEV planners are trained on NAVSIM (nuPlan) data, as is AutoVLA’s released checkpoint. SimLingo is trained in CARLA and Alpamayo-1.5 on NVIDIA’s PhysicalAI-AV recordings. None is fine-tuned here, so every policy is read on a corpus and sensor rig it was not trained on. For the BEV planners the nuScenes front image is centre-cropped to the 4:1 aspect of NAVSIM’s stitched front view with the sky removed and resized to the network’s input size, the logged speed and acceleration fill the status vector, and DiffusionDriveV2 receives the LiDAR sweep. The VLA policies receive the nuScenes cameras mapped approximately onto the views they expect. DiffusionDrive’s diffusion seed is fixed so that paired queries share their noise.

The primary corpus is nuScenes [1] keyframes: corridor pedestrians (lateral clearance to the ego corridor ≤ 1 m) at 5–12 m, their off-road counterparts, and 30 approach sequences at 5.8–14.6 m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive). Alpamayo-1.5 covers 170 of the 236, needing 1.6 s of history and 6.4 s of future within the scene. Section IV-B repeats $\mathcal{E}_R$ and $\mathcal{E}_N$ on a second, right-hand-drive corpus—the NAVSIM test split [2], 254 Singapore scenes with a pedestrian near the ego corridor, of which 246 carry a complete logged pedestrian future—used only for the transfer and decomposition results. Table I is measured on the nuScenes frames alone.

B. The Two Axes

The diagnosis reads a policy on the two axes of (3). We apply both interventions to the 246 NAVSIM Singapore near-pedestrian scenes and read, per scene, F , the plan displacement caused by removing the target pedestrian, and I , that caused by the night-style perturbation.

a) When the plan does move, which way does it move?: F measures how far a plan moves, not where it moves to. We therefore decompose it (Fig. 2d), reading F against I of (3)—the same policy’s jitter floor under an unrelated edit—and against ΔS , the change in clearance to the pedestrian’s logged future. A response counts as avoidance only if all three hold: the plan moved ($F \geq 0.5$ m), it moved more than the policy’s own jitter ($F > I$), and the movement bought distance ($\Delta S \geq 0.5$ m).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of what F registers sits inside the policy’s own variability—below the $F = I$ line of Fig. 2d. Second, avoidance is rare where

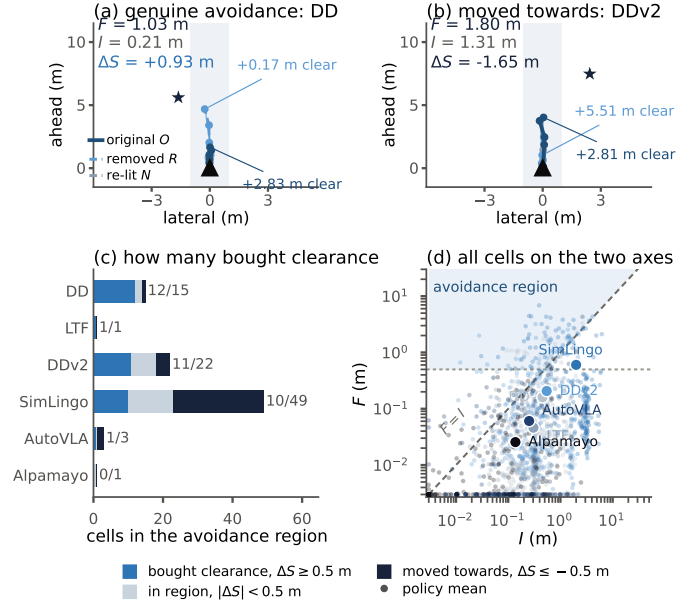


Fig. 2. The two axes. (a) Genuine avoidance: with the pedestrian in view (X^O , solid) DiffusionDrive stays short of its logged future (dashed from the star), while blind to it (X^R , dashed) it reaches it. X^N gives I . (b) Moved towards: DiffusionDriveV2 plans further with the pedestrian than without. (c) Cells in the avoidance region ($F \geq 0.5$ m, $F > I$) per policy, by ΔS : green ≥ 0.5 m, red ≤ -0.5 m, grey between. Label: green over all. (d) All 246 Singapore cells on the axes of (3), by policy, with means.

it is called for. The logged ego is stationary through most of this corpus, so a plan that does not move is usually the correct one. We therefore replay every scene at input speeds of 4 and 8 m/s, as in Section III, and read only the cells in which the pedestrian-blind plan does come within 1 m of the pedestrian’s logged future. Across 6 policies that is 787 of 4428 cells (11–30% per policy), and 15 of them—1.9%—are genuine avoidance. On those cells the median clearance changes by 0.00–0.03 m and the median planned distance by -0.02 – 0.08 m: facing a pedestrian it would otherwise reach, no policy’s plan diverts or slows under our open-loop protocol. Third, the displacement that does occur is not aimed: of the 121 cells in which the plan moved, 43 increased the clearance and 42 decreased it. Per policy the split differs sharply (Fig. 2c): DiffusionDrive reacts in few scenes but in the right direction (median $\Delta S = 2.16$ m), while SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median -0.59 m), and it also has the highest rate of sub-1.5 s time-to-proximity (Table I). SimLingo is trained in simulation and run here on real images, so part of what we measure is a sim-to-real gap, which is exactly what a check-up at a new site should surface. Finally, what a policy yields does not grow with the hazard: the rank correlation between ΔS and the deceleration the scene demands is between -0.37 and 0.07 .

An exam that rewarded responsiveness alone would therefore rank first the policy with the most sub-1.5 s encounters. A score merges these three readings into one

number. The edited frames keep them apart.

C. The Five Exams

Table I gives the six profiles, read under our open-loop protocol on the 236 near-pedestrian nuScenes frames: the behaviour behind the ranks of Section IV-E.

a) Exposure: The six policies differ first in how far they plan, and this alone will turn out to explain more of the collision picture below than perception does: exposure spans a factor of four, from DiffusionDrive at half the human’s distance to SimLingo at more than twice it, with AutoVLA and Alpaymayo-1.5 near the human’s own.

b) Hazard sensitivity: On these frames, no policy’s plan responds to the pedestrian: on units that need a reaction, seeing it buys only 0.00–0.10 of safety on a $[-1, 1]$ scale, against 0.57 when the logged human trajectory H takes the place of X^O in (7) (Table I). The benefit is also unreliable in direction: counting only cases where seeing the pedestrian changes the outcome, it turns a collision into a clean pass about as often as it turns a pass into a collision, for every policy (DiffusionDrive 9 vs. 5, SimLingo 18 vs. 30, the rest at most 5 either way, all McNemar $p \geq 0.11$) — what reads as a safety margin in Table I does not carry over to the individual case.

c) Scaling: The benefit does not grow with the hazard: its rank correlation with a_{req} is negative for 5 policies, down to -0.28 for DiffusionDrive. DiffusionDriveV2 is the only exception (0.23), and its benefit stays below 0.15. Replacing the hazard axis by the time-based TTC_0 , whose sign is reversed by construction, leaves the picture unchanged (-0.07 to 0.32 , against -0.28 to 0.23). The human reference is 0.13: a driver who already yields at every hazard level has little room left to scale, so this exam is informative where HS is small, which is where all six policies sit.

d) Specificity: A policy whose plan moves when nothing requires it is not cautious but unsteady, and steadiness is what turns out to travel across a domain shift (Section IV-E). SimLingo’s plan moves most when nothing requires it ($\text{SP} = 0.44$), while the other five stay steadier, at 0.65 – 0.84 .

e) Lighting: Removing the pedestrian moves every plan less than the night-style perturbation does ($\text{CFR} < 1$ for all six, Table I). Section IV-D reads this exam in depth.

f) Collisions: Collisions with the pedestrian’s logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpaymayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the lowest specificity.

g) Why the risk goes unmanaged: Two observations on the three BEV planners explain the diagnosis. Their

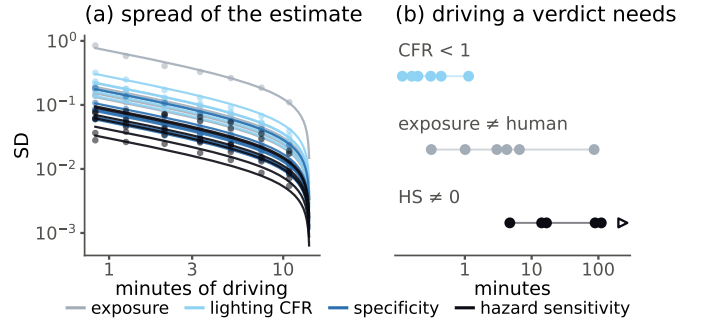


Fig. 3. What a verdict costs. One frame is 5 s of near-pedestrian driving. (a) Spread of each estimate against sample size with its fitted law, one line per policy. (b) Driving needed per verdict, one dot per policy, off-scale values at the axis end.

BEV semantic heads do not decode the pedestrian: on 520 NAVSIM near-pedestrian scenes the pedestrian channel never wins the argmax inside a pedestrian’s box. And their plans end 5–9 m ahead while the median corridor pedestrian stands 20 m away, so collisions rise with input speed identically with and without the pedestrian in view (Table I). The limit is geometric before it is perceptual.

D. The Lighting Exam in Depth

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. It is the verdict that transfers between driving sides for every policy (Section IV-E).

a) Magnitude: Removing a pedestrian from the corridor moves every policy’s plan only 0.23–0.47 times as far as the night-style perturbation does, with every 95% interval below one (Table I). On detector-verified edits $\text{CFR} = 0.30$ – 0.64 .

b) Direction: faster, and therefore closer: Three policies plan faster under the night-style perturbation, DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%, and their plans pass 0.24–0.47 m closer to the pedestrian, while DiffusionDriveV2, the only policy that also reads LiDAR, slows (-8%) and keeps more distance. Whether these profiles and orderings persist across a substantial domain shift is tested in Section IV-E.

E. Prognosis: What Transfers, and at What Cost

Two questions decide whether a few hundred edited frames can stand in for a rank: which verdicts carry over to the other driving side, and how many frames each verdict needs.

a) A pre-registered transfer test: Does a diagnosis made on one driving side describe the same policy on the other? Before computing any right-hand-drive statistic, we fixed six predictions, P1–P6 in Table II, and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table II). The registration covered the six policies public at that time. NVIDIA subsequently released Alpaymayo 1.5, which

TABLE I

The full report for the six policies. Rows are readings, columns policies. Brackets: 95% CI. Arrows in cells: outside the reference threshold of Section III; bold: best per row, or $p < 0.01$ for the two rows of N . Speed and clearance of N : change under the night-style perturbation, on the same frames (negative clearance = closer). Human: the logged trajectory scored the same way against each policy’s blind plan. Standard scores: nuScenes L2 on the same frames, NAVSIM EPDMS on all 783 Singapore scenes and on the 260 near-pedestrian ones, and the share of moving-ego scenes with pedestrian TTC < 1.5 s per driving side.

Reading \ Policy	DD	LTF	DDv2	SimLingo	AutoVLA	Alpamayo 1.5	Human
Exams (236 nuScenes frames)							
Exposure (ratio to human) ≈ 1	0.50↓	0.79↓	0.88	2.19↑	1.01	1.13	1.00
95% CI	[0.46, 0.55]	[0.75, 0.84]	[0.77, 0.95]	[1.98, 2.65]	[0.97, 1.11]	[1.06, 1.21]	
Hazard sens. HS $\in [-1, 1]$ ↑	0.06↓	0.03↓	0.10↓	0.00↓	0.02↓	0.01↓	0.57
95% CI	[-0.01, 0.14]	[0.00, 0.06]	[0.05, 0.16]	[-0.03, 0.03]	[-0.00, 0.05]	[-0.01, 0.02]	
Scaling (rank corr.) ↑	-0.28↓	-0.13↓	0.23	-0.22↓	-0.18↓	-0.23↓	0.13
95% CI	[-0.50, -0.05]	[-0.34, 0.07]	[-0.02, 0.50]	[-0.38, -0.08]	[-0.34, 0.01]	[-0.49, 0.03]	
Specificity SP $\in (0, 1]$ ↑	0.76↓	0.78↓	0.65↓	0.44↓	0.76↓	0.84	–
95% CI	[0.73, 0.80]	[0.75, 0.81]	[0.61, 0.70]	[0.38, 0.50]	[0.72, 0.79]	[0.81, 0.87]	
Lighting CFR (ratio) ↑	0.23↓	0.23↓	0.47↓	0.23↓	0.44↓	0.25↓	–
95% CI	[0.17, 0.30]	[0.19, 0.28]	[0.37, 0.58]	[0.18, 0.29]	[0.33, 0.55]	[0.16, 0.35]	
Collision at 8 m/s, vis. / rm. (%) ↓	7.6 / 9.2	9.9 / 10.7	3.1 / 6.1	40.5 / 36.6	32.1 / 29.8	11.2 / 11.2	–
speed of N (%) ↓	37%	12%	−8%	16%	−0.3%	−2%	–
clearance of N , $\Delta_N S$ (m) ↑	−0.37	−0.24	0.15	−0.47	0.005	0.03	–
Standard scores (rank)							
L2 (m), vis. / rm. ↓	0.98 / 0.99	0.88 / 0.91	1.35 / 1.31	3.28 / 3.35	0.68 / 0.66	0.77 / 0.72	–
EPDMS, all scenes ↑	0.535 (4)	0.509 (5)	0.652 (3)	0.359 (6)	0.683 (1)	0.663 (2)	–
EPDMS, near-ped. ↑	0.744 (2)	0.735 (3)	0.367 (6)	0.583 (4)	0.413 (5)	0.760 (1)	–
TTC < 1.5 s (%), LHD ↓	3.2 (1)	5.4 (2)	18.3 (3)	61.3 (6)	19.4 (4)	25.4 (5)	–
TTC < 1.5 s (%), RHD ↓	14.1 (1)	16.5 (2)	22.4 (5)	56.5 (6)	21.2 (4)	20.0 (3)	–

TABLE II

Pre-registered transfer test. Predictions fixed on 88 Boston frames, tested on 148 Singapore frames; five policies, the replaced sixth changes no verdict (Section IV-E).

No.	Prediction	Holds	Evidence
P1	Lighting outweighs the pedestrian (CFR < 1) for every policy	yes	max upper CI 0.70
P2	Hazard sensitivity stays below 0.15 (upper CI)	no	DD, DDv2 above
P3	Exposure ordering transfers ($\rho \geq 0.6$; SimLingo top, DD bottom)	yes	$\rho = 0.90$
P4	Specificity ordering transfers ($\rho \geq 0.6$; SimLingo lowest SP)	yes	$\rho = 0.60$
P5	Point values stay within the left-hand CI ($\geq 70\%$ of 15 cells)	no	60%
P6	SimLingo collides most, visible vs. removed within 3 pts	yes	gap 2.7 pts

accepts navigation instructions, and we use it in place of Alpamayo-R1 throughout this paper. Because both driving sides of Alpamayo 1.5 were measured together, its numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the six verdicts.

b) What transferred: 4 of the six predictions hold: the lighting verdict, the two orderings and the collision outcome. The night-style perturbation outweighs the pedestrian in every policy. Exposure keeps its order ($\rho = 0.90$), and so does specificity, with SimLingo the lowest. SimLingo collides most, with the visible and removed rates within 3 points of each other.

c) What did not: Two predictions fail.

- Point values drift: only 60% of the model–exam cells stay inside their Boston interval.

- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity (HS = 0.09 and 0.08).

d) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

e) The standard scores on the same frames: Running the standard scores on the same six policies (Table I) shows what a rank cannot see. Removing the pedestrian changes none of the nuScenes open-loop numbers, the L2 error by at most 0.07 m and the collision rates by at most 1.7 points, so the metric cannot tell whether a policy saw the pedestrian, and its ordering follows how far a policy’s planned distance departs from the human’s (0.66 with [exposure − 1]). On NAVSIM the leaderboard ordering over all 783 Singapore scenes is unrelated to the ordering on the 260 near-pedestrian scenes ($\rho = -0.09$). Its first, AutoVLA, falls to fifth and its third, DiffusionDriveV2, to last, because the full-set score is carried by the map-compliance gates (95% of the spread) and none of the exams above is visible in it. Pedestrian TTC, by contrast, orders the policies the same way on both driving sides ($\rho = 0.77$) and coincides with exposure (1.00, 0.77).

f) Which exam travels: This is the question a deployment decision has to answer. The target is the ordering of the six policies on the Singapore near-pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score,

correlate with it at -0.54 to 0.49 . The same pedestrian-centred readouts taken on the other driving side do (0.83 – 1.00). Hazard sensitivity and lighting invariance measure something else and do not predict these orderings. Of the five exams, the one whose ordering best matches how often a policy keeps clear of the pedestrian on the new side (share of scenes with minimum time-to-proximity ≥ 1.5 s, the complement of the TTC column of Table I) is not the one that measures how much a policy reacts but the one that measures how steadily it drives: over the six policies and the 246-scene pool of Section IV-B, specificity gives $\rho = 0.94$, against -0.77 for exposure and -0.37 for CFR (higher is better for all three exams, so a positive ρ means the exam’s ordering carries over). This exam was not pre-registered, and we report it as exploratory. Cross-policy correlations rest on six policies, and we read them as orderings, not as tests. Specificity asks how far a plan moves when an irrelevant part of the scene is edited. A policy that answers “not at all” drives steadily, and steadiness is what survives the change of side. The decomposition of Section IV-B says why: displacement that is not aimed away from the pedestrian is not protection, and a policy producing a lot of it comes close to the pedestrian on both sides. A rank does not carry the near-pedestrian ordering to a new site. The exposure and specificity exams, read from the same frames, do.

g) What a verdict costs: For every exam and policy we subsample n of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with $\text{sd}(n) = a\sqrt{1/n - 1/M}$, $M = 170$ (Fig. 3).

h) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ($R^2 = 0.97$ – 1.00). The fitted a converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian ($\text{CFR} < 1$) is settled for every policy within 14 frames.
- Exposure differs from the human’s is settled within 51 frames for the four policies whose exposure is far from one. For AutoVLA and Alpamayo-1.5 it would take hundreds to thousands of frames, which is itself a finding, since they plan a human-like distance.
- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach $\rho \geq 0.6$ with the full ordering) and specificity with 40 (99%).

i) Poorly fitting exams: Hazard sensitivity behaves differently. Its fits are poor (R^2 as low as 0.26) wherever few units need a reaction. Deciding that $\text{HS} \neq 0$ needs 57–1,326 frames, except for SimLingo, whose estimate sits on the threshold, and the ordering reaches 84% stability only at 90 frames.

The division is therefore clear. A few dozen frames, about 3 logged scenes, diagnose exposure, specificity and lighting. Whether a policy manages hazard needs hundreds of frames that contain a genuine need to react.

V. Discussion

a) Limitations: The exams are open-loop, replayed from logs and re-anchored to the logged pose. The night-style perturbation is photometric, not captured. The counterfactual pairs are edited, not captured: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA policies are edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames, SimLingo’s 2.5 s plans are extrapolated to NAVSIM’s 4 s, and cross-policy correlations rest on six policies.

b) Next steps: The check-up is the first half of a benchmark-to-deployment loop. The paired data will next be captured rather than constructed, the same scene with and without a pedestrian recorded on our vehicle and re-rendered with 3D Gaussian splatting. The day–night pairs will move from a photometric transform to captured or physically rendered lighting. And the VLA policies will be post-trained on target-domain data and re-examined, so that the diagnosis measures what post-training recovers. The sample-size laws above price that collection.

VI. Conclusion

A benchmark score reports an outcome, not the behaviour behind it, so a rank cannot be trusted to hold at a new site. The check-up reads that behaviour from a few hundred logged frames through verified, pixel-level counterfactual edits, reduced to two causal axes, F and I , on which five exams are built. Three results carry the paper. On the cells in which a pedestrian lies on the planned path, only 1.9% of responses are genuine avoidance and, in open loop, no policy’s plan diverts or slows, while the night-style perturbation moves every plan further than the pedestrian does. Diagnosed on one driving side and tested on the other, orderings and verdicts carry over and point values do not, and specificity orders the policies as their clearance on the new side does ($\rho = 0.94$) where the standard scores reach only -0.54 to 0.49 . Each verdict is priced in advance, lighting within 14 frames and exposure within 51, so a deployment decision costs tens of target-domain frames rather than a fleet test.

References

- [1] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, “nuScenes: A multimodal dataset for autonomous driving,” in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
- [2] D. Dauner, M. Hallgarten, T. Li, X. Weng, Z. Huang, Z. Yang, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, “NAVSIM: Data-driven non-reactive autonomous vehicle simulation and benchmarking,” in Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2024.
- [3] S. Z. Zhao, L. Wang, H. Ruan, Y. Bao, Y. Chen, Z. Leng, A. Ravichandran, H. He, Z. Zhou, X. Han, A. Peri, Z. Huang, P. Desai, H. Christensen, J. Ma, and B. Zhou, “BridgeSim: Unveiling the OL–CL gap in end-to-end autonomous driving,” arXiv preprint arXiv:2604.10856, 2026.

- [4] Z. Li, Z. Yu, S. Lan, J. Li, J. Kautz, T. Lu, and J. M. Alvarez, "Is ego status all you need for open-loop end-to-end autonomous driving?" in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [5] F. Codevilla, E. Santana, A. M. López, and A. Gaidon, "Exploring the limitations of behavior cloning for autonomous driving," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [6] K. D. Kusano, J. M. Scanlon, Y.-H. Chen, T. L. McMurry, T. Gode, and T. Victor, "Comparison of Waymo rider-only crash rates by crash type to human benchmarks at 56.7 million miles," *Traffic Injury Prevention*, vol. 26, no. S1, pp. S8–S20, 2025.
- [7] Physical Intelligence, A. Amin, R. Aniceto, A. Balakrishna, K. Black, K. Conley, G. Connors, J. Darpinian, K. Dhabalia, J. DiCarlo, D. Driess, M. Equi, A. Esmail, Y. Fang, C. Finn, C. Glossop, T. Godden, I. Goryachev, L. Groom, H. Hancock, K. Hausman, G. Hussein, B. Ichter, S. Jakubczak, R. Jen, T. Jones, B. Katz, L. Ke, C. Kuchi, M. Lamb, D. LeBlanc, S. Levine, A. Li-Bell, Y. Lu, V. Mano, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, C. Sharma, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, W. Stoeckle, A. Swerdlow, J. Tanner, M. Torne, Q. Vuong, A. Walling, H. Wang, B. Williams, S. Yoo, L. Yu, U. Zhilinsky, and Z. Zhou, " $\pi_{0,6}^*$: A VLA that learns from experience," *arXiv preprint arXiv:2511.14759*, 2025.
- [8] N. Kalra and S. M. Paddock, "Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability?" *RAND Corporation*, Tech. Rep. RR-1478, 2016.
- [9] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *International Conference on Machine Learning (ICML)*, 2015.
- [10] I. Gulrajani and D. Lopez-Paz, "In search of lost domain generalization," in *International Conference on Learning Representations (ICLR)*, 2021.
- [11] N. Kachaev, M. Kolosov, D. Zelezetsky, A. K. Kovalev, and A. I. Panov, "Don't blind your vla: Aligning visual representations for ood generalization," *arXiv preprint arXiv:2510.25616*, 2025.
- [12] A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, R. Ren, A. Pan, X. Yin, M. Mazeika, A.-K. Dombrowski et al., "Representation engineering: A top-down approach to AI transparency," *arXiv preprint arXiv:2310.01405*, 2023.
- [13] E. Todd, M. L. Li, A. S. Sharma, A. Mueller, B. C. Wallace, and D. Bau, "Function vectors in large language models," *arXiv preprint arXiv:2310.15213*, 2024. [Online]. Available: <https://arxiv.org/abs/2310.15213>
- [14] K. Chitta, A. Prakash, B. Jaeger, Z. Yu, K. Renz, and A. Geiger, "TransFuser: Imitation with transformer-based sensor fusion for autonomous driving," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [15] B. Liao, S. Chen, H. Yin, B. Jiang, C. Wang, S. Yan, X. Zhang, X. Li, Y. Zhang, Q. Zhang, and X. Wang, "DiffusionDrive: Truncated diffusion model for end-to-end autonomous driving," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [16] J. Zou, S. Chen, B. Liao, Z. Zheng, Y. Song, L. Zhang, Q. Zhang, W. Liu, and X. Wang, "DiffusionDriveV2: Reinforcement learning-constrained truncated diffusion modeling in end-to-end autonomous driving," *arXiv preprint arXiv:2512.07745*, 2025.
- [17] K. Renz, L. Chen, E. Arani, and O. Sinavski, "SimLingo: Vision-only closed-loop autonomous driving with language-action alignment," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [18] Z. Zhou, T. Cai, S. Z. Zhao, Y. Zhang, Z. Huang, B. Zhou, and J. Ma, "AutoVLA: A vision-language-action model for end-to-end autonomous driving with adaptive reasoning and reinforcement fine-tuning," *arXiv preprint arXiv:2506.13757*, 2025.
- [19] NVIDIA, "Alpamayo-R1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail," *arXiv preprint arXiv:2511.00088*, 2025.
- [20] —, "Alpamayo 1.5: An open reasoning vision-language-action model with navigation guidance," 2026, model card and code, <https://github.com/NVlabs/alpamayo1.5>.
- [21] B. Jaeger, K. Chitta, and A. Geiger, "Hidden biases of end-to-end driving models," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [22] X. Jia, Z. Yang, Q. Li, Z. Zhang, and J. Yan, "Bench2Drive: Towards multi-ability benchmarking of closed-loop end-to-end autonomous driving," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.
- [23] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, "Beyond accuracy: Behavioral testing of NLP models with CheckList," in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [24] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019.
- [25] P. de Haan, D. Jayaraman, and S. Levine, "Causal confusion in imitation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [26] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [27] P. Jacob, É. Zablocki, H. Ben-Younes, M. Chen, P. Pérez, and M. Cord, "STEEEX: Steering counterfactual explanations with semantics," in *European Conference on Computer Vision (ECCV)*, 2022.
- [28] R. Roelofs, L. Sun, B. Caine, K. S. Refaat, B. Sapp, S. Ettinger, and W. Chai, "CausalAgents: A robustness benchmark for motion forecasting using causal relationships," *arXiv preprint arXiv:2207.03586*, 2022.
- [29] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, "Segment anything," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [30] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky, "Resolution-robust large mask inpainting with fourier convolutions," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [31] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [32] E. T. Hall, *The Hidden Dimension*. Doubleday, 1966.
- [33] D. Gettman and L. Head, "Surrogate safety measures from traffic simulation models," *Transportation Research Record*, vol. 1840, no. 1, pp. 104–115, 2003.